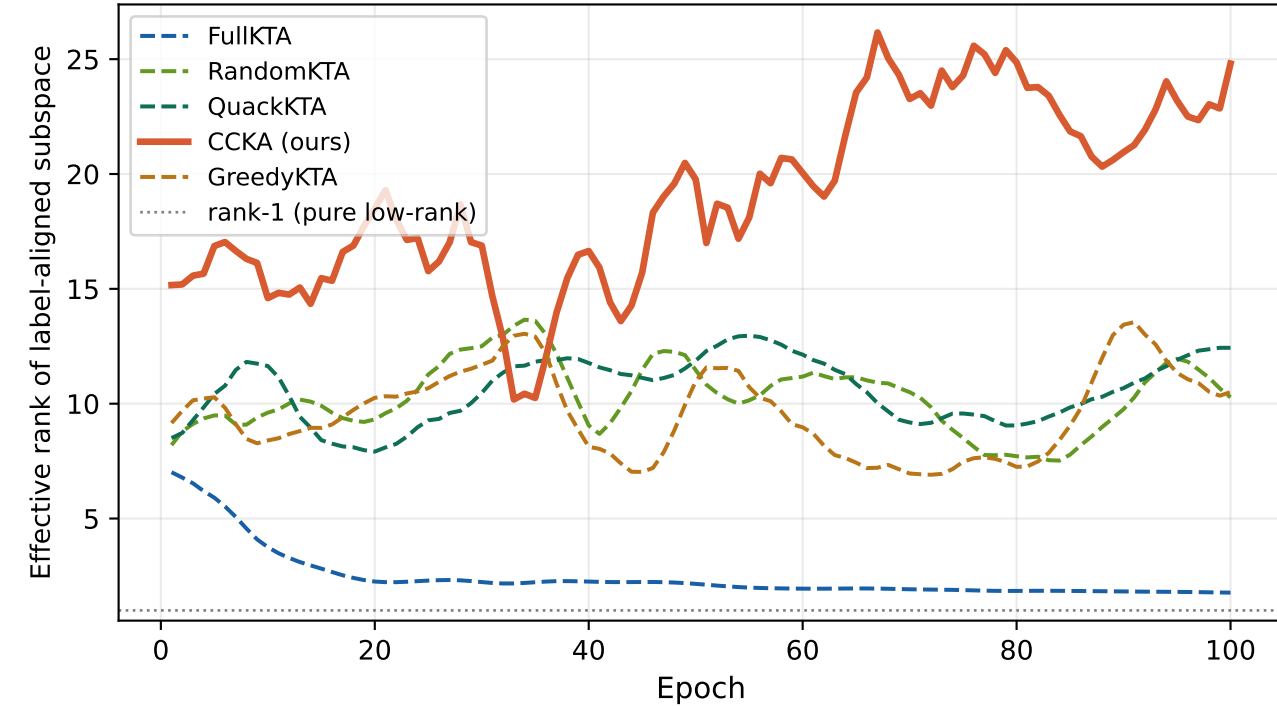


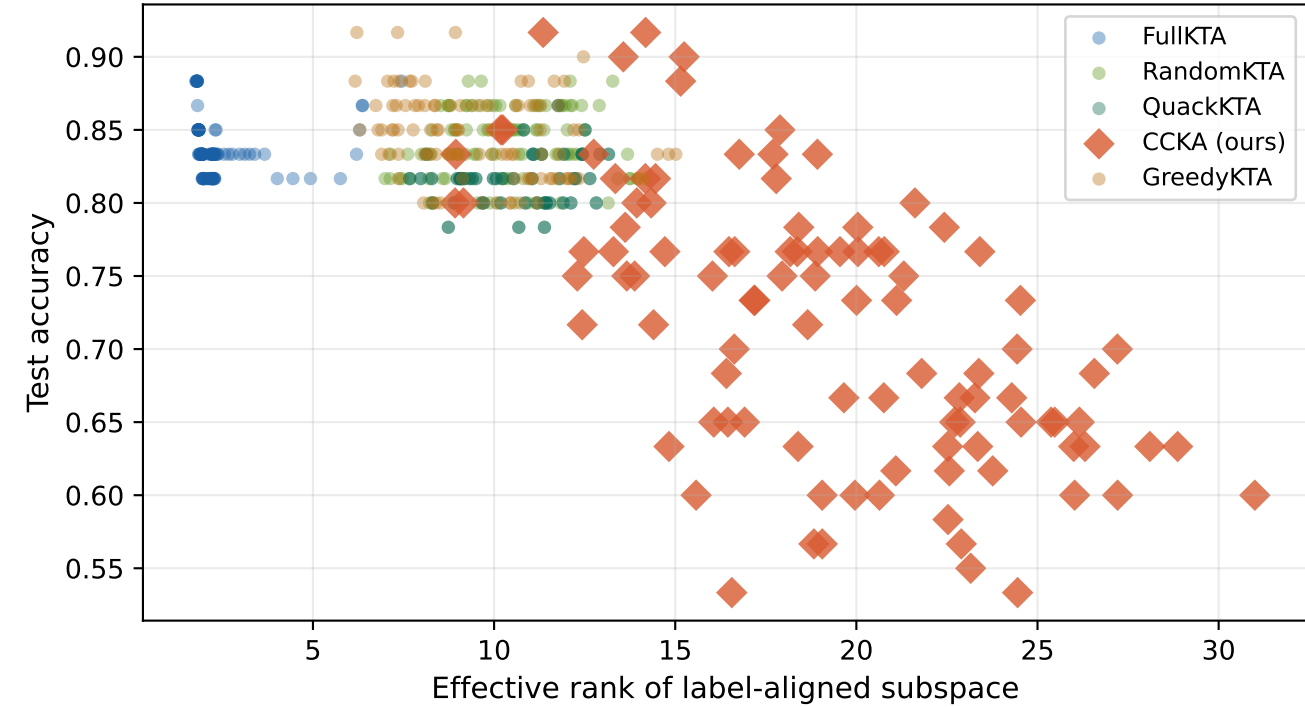
Spectral / Effective-Rank Analysis — DONUTS

Testing the low-rank discriminative alignment theory of CCKA

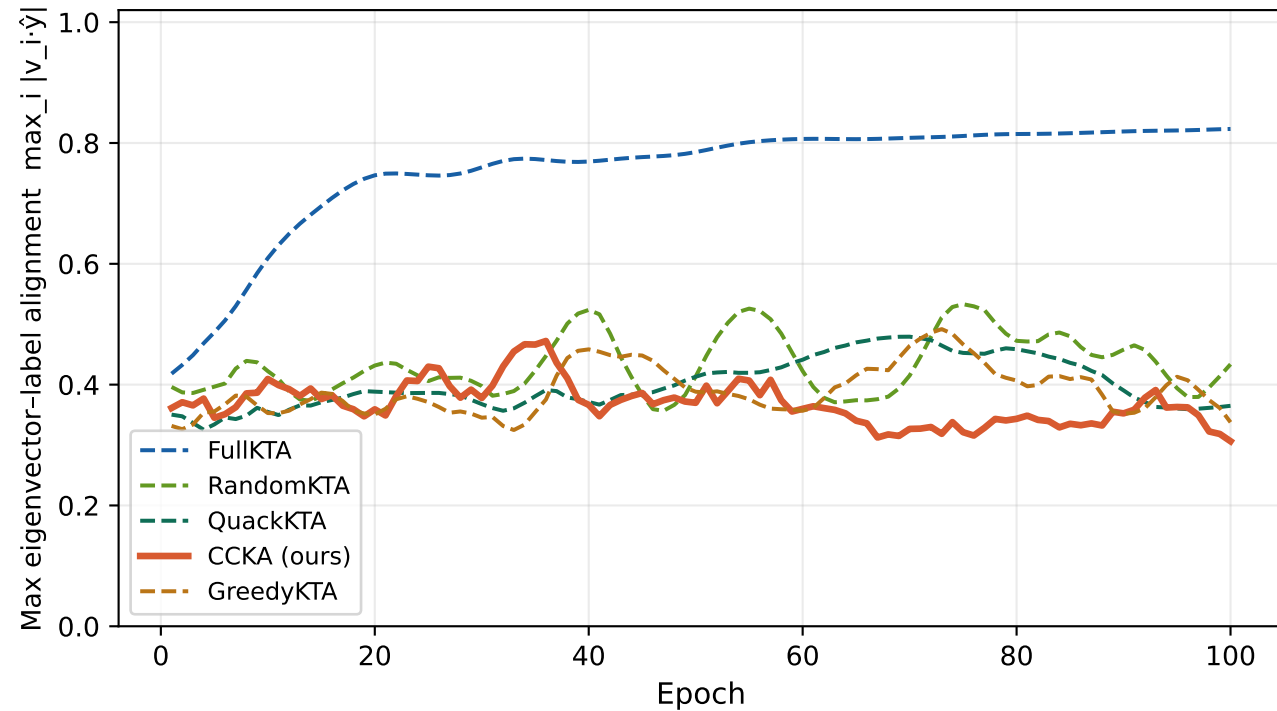
Effective rank over training — donuts
CCKA expected near 1 (rank-1); FullKTA much higher



Effective rank vs accuracy — donuts
CCKA (◆) achieving high accuracy at LOW eff-rank = rank-1 sufficiency



Best-aligned eigenvector over training — donuts
CCKA collapse (~ep64) should show a sharp drop here



KTA decomposition by rank — donuts
Near 1.0 = essentially all KTA in the top 1-2 directions (low-rank)

